An Introduction to Applied Bayesian Statistics with Stan

2026-08-17

Overview

- **Goals**

  Provide the foundation and orientation so that you can

  - Use Stan (or **brms**) in your work
  - Understand and enjoy StanCon talks!

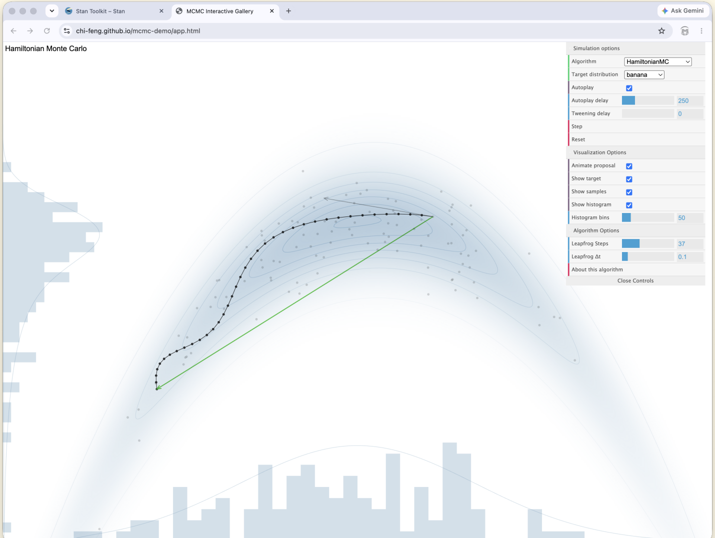
- **Outline**
  - About Stan ✓
  - Bayesian models and workflow ✓
  - Model specification in Stan ✓
  - Multilevel model specification in **brms** ✓
  - HMC and multi-level models ←

- **Slides** stancon_2026_stan_intro

Hamiltonian Monte Carlo

Hamiltonian Monte-Carlo

MCMC Interactive Gallery - https://chi-feng.github.io/mcmc-demo/app.html



Hamiltonian Monte-Carlo

- Generate draws from the **joint** log posterior density $\log p(\theta, y)$
  - follow the **Hamiltonian flow** according to the **gradient** of the density
  - Stan's automatic-differentiation computes these gradients
- HMC computes on the unconstrained scale $\mathbb{R}^N$
  - Stan provides constraining/unconstraining transforms
- The Hamiltonian flow is continuous; discretized to **steps**
  - One draw from the posterior consists of many steps
- HMC parameters must be tuned for the posterior geometry
  - Tune step size and estimate mass matrix during warmup, determine number of steps on-the-fly (NUTS)
- See Michael Betancourt's excellent A Conceptual Introduction to Hamiltonian Monte Carlo

Euclidean Hamiltonian Monte Carlo

- **Phase space**: q position (parameters); p momentum
- **Target density**: $\pi(q)$
- **Potential energy**: $V(q) = -\log \pi(q)$
- **Metric (mass matrix)**: M
  - rescales and rotates the dynamics to match posterior geometry
- **Momentum refresh**: $p \sim \text{MultiNormal}(0, M)$
- **Kinetic energy**: $T(p) = \frac{1}{2} p^\top M^{-1} p$
- **Hamiltonian**: $H(p, q) = V(q) + T(p)$

$$\frac{dq}{dt} = M^{-1} p \qquad\qquad \frac{dp}{dt} = \nabla_q \log \pi(q)$$

Leapfrog integrator

- Solves Hamilton's equations by **simulating dynamics**

  (symplectic [volume preserving]; ϵ^3 error per step, ϵ^2 total error)

- Given: **step size** ϵ, **mass matrix** M, **parameters** q
- **Initialize kinetic** energy, $p \sim \mathrm{Normal}(0, \mathbf{I})$
- **Repeat** for L leapfrog steps:

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad \text{[half step in momentum]}$$

$$q \leftarrow q + \epsilon \, M^{-1} \, p \quad \text{[full step in position]}$$

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad \text{[half step in momentum]}$$

HMC Tuning Parameters

- HMC simulate Hamiltonian trajectory of a particle through parameter space
  - at each iteration, provide a refresh of random momentum ρ where

    $$\rho \sim \mathrm{MultiNormal}(0, M),$$

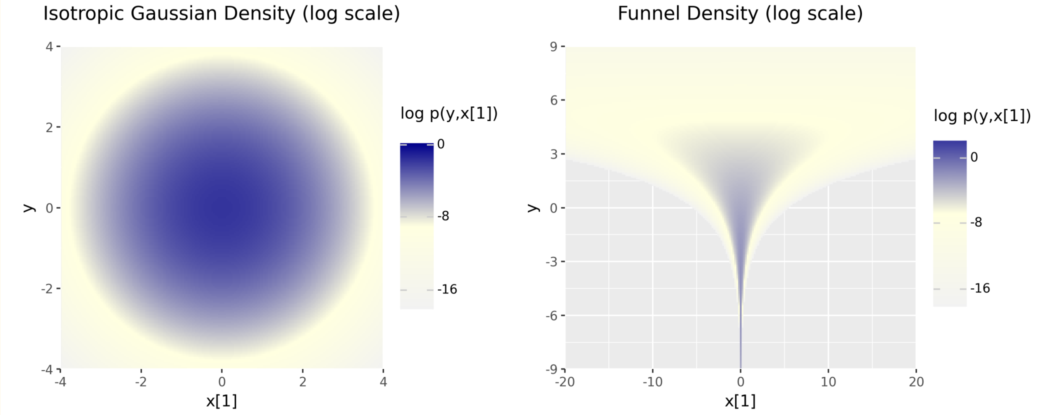
    where M is the positive definite mass matrix
  - this puts posterior directions on comparable scales

    (we have a high-dimensional posterior)
- The well-tuned HMC sampler requires that:
  - one trajectory is composed of just enough steps
  - the step-size just small enough to follow the Hamiltonian
  - the mass matrix is a good *linear pre-conditioner*

Efficiency Tuning

- The HMC sampler explores the posterior density of the *parameter space*
- If the posterior is (approximately) multivariate normal the mass matrix produces an isotropic Gaussian posterior - a single step in every direction is accurate
- If the posterior is sharply curved, no single mass matrix is valid everywhere
  - during warmup the step-size is adjusted smaller and smaller
  - the sampler fails to make progress
- We can rewrite Stan programs in terms of *transformed parameters*
  - see the Stan User's Guide Efficiency Tuning chapter.
- We have already seen this: the centered and non-centered parameterizations of a hierarchical model

Efficiency Tuning for Hierarchical Models



Easy vs. hard posterior densities

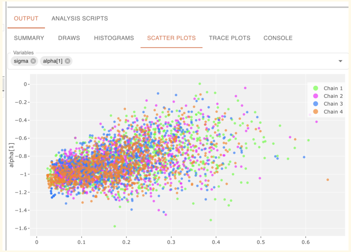
Efficiency Tuning for Hierarchical Models

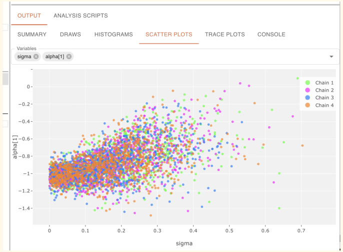
- The isotropic Gaussian density is *not* a hierarchical model
  - it is the product of N normal distributions
- The funnel distribution is hierarchical
  - the scale of elements x depends on global parameter y
- **But** *the posterior density is the product of the prior times the likelihood*
  - shape of the posterior morphs as the model sees more data
- You see this already in the Stan Playground scatter plots

Baseball Example Posterior Densities

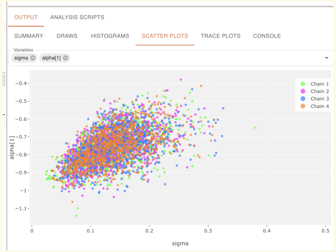
Centered parameterization | **Non-centered parameterization**

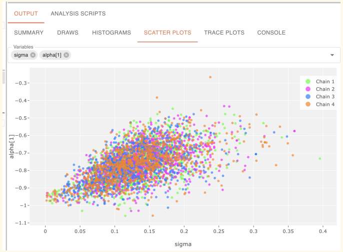
Low





Higher





Centered vs. Non-Centered Parameterization: the Data

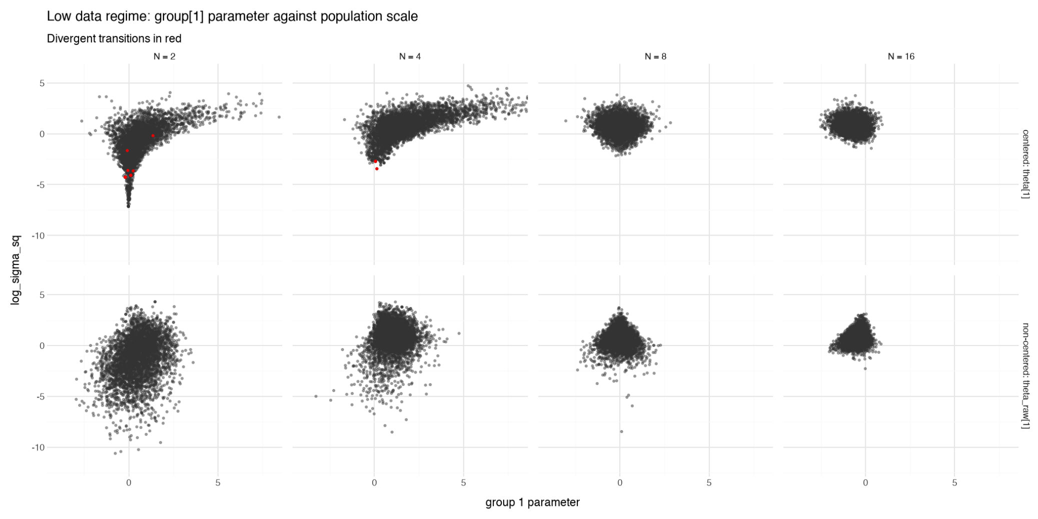
- The R-hat statistic and ESS are related
  - see Nested R-hat: Assessing the convergence of Markov chain Monte Carlo when running many short chains
  - a further consequence is slow sampling time (low ESS/s) as well.
- In low data regimes the non-centered parameterization improves performance
  - better ESS
  - faster total sampling time
- In high data regimes the centered parameterization wins
- The baseball dataset:
  - "low data" and "more data"
  - **is not generated according to the model**

Centered vs. Non-Centered Parameterization: Simulated Data

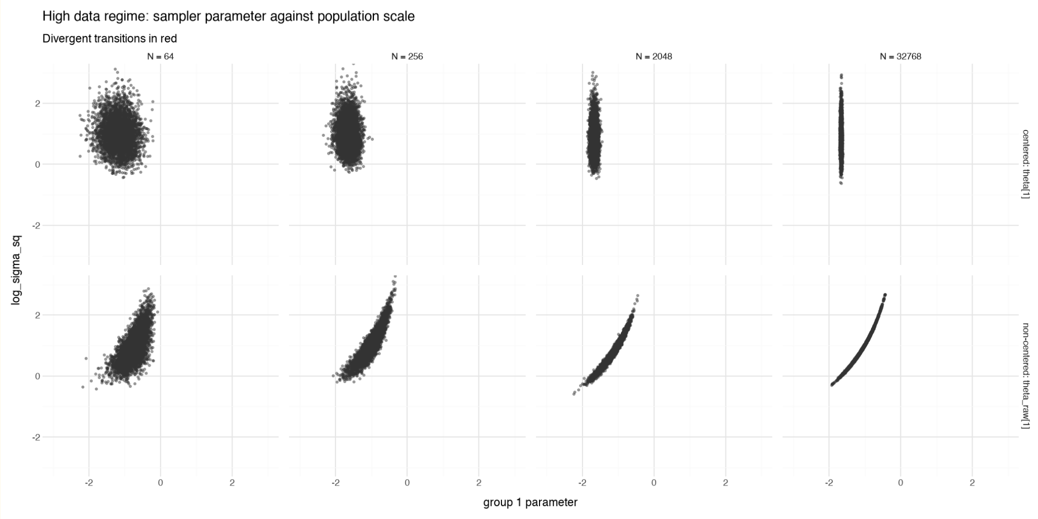
```
data {
  int<lower=0> N_groups, N_obs;
}
generated quantities {
  vector[N_groups] theta;
  for (i in 1:N_groups) {
    theta[i] = std_normal_rng();  // mu = 0, sigma = 1
  }
  vector[N_groups] p = inv_logit(theta);  // probability in range (0,1)
  array[N_obs, N_groups] int<lower=0, upper=1> y;
  for (i in 1:N_obs) {
    for (j in 1:N_groups) {
      y[i, j] = bernoulli_rng(p[j]);
    }
  }
}
```

- Fit to subsets of the data sizes $0, 2, 4, 8, 10, 100, 1000, 10000$

Centered vs. Non-Centered Parameterization: Small Data



Centered vs. Non-Centered Parameterization: Big Data



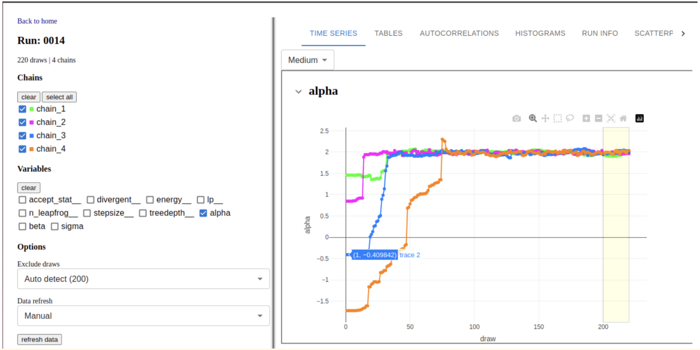
Centered vs. Non-Centered Parameterization: Complicated Data

- These plots were generated from synthetic data generated according to the model
- Beyond a few simple cases, there is no one good solution
- *Note:* **brms** always uses the non-centered parameterization
- Some References
  - A General Framework for the Parametrization of Hierarchical Models
    Papaspiliopoulos et al, 2007
  - Hamiltonian Monte Carlo for Hierarchical Models
    Betancourt and Girolami, 2013
  - The Stan User's Guide chapters:
    Problematic Posteriors, Reparameterizations Efficiency Tuning

Diagnosing Complications: the MCMC monitor

MCMC Monitor : https://github.com/flatironinstitute/mcmc-monitor

MCMC Monitor enables tracking and visualization of MCMC processes executed with Stan in local or remote web browsers. When you run a sampler, you can configure Stan to generate output to a directory on your computer. MCMC Monitor reads this output and displays it in the web app, with real-time updates. As you track the progress of the run, MCMC provides diagnostic plots and statistics.



Questions and Discussion

Stan Documentation

- Stan Documentation Suite
  - Stan User's Guide
  - Reference Manual - just the facts
  - Functions Reference - Stan math library functions available in the Stan language
- Learning Resources
  - Tutorials - just a few of many
  - Stan Case Studies - best practices, deep dives for many models
  - StanCon Talks - past StanCons
  - Field Guides - education, ecology, epidemiology, cogsci, pharma
- Stan Forums - your questions answered